

Diabetes Risk Prediction, Fairness Evaluation, and Causal Analysis Using Propensity Score Matching

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Abstract— Diabetes is a major public health concern associated with clinical, behavioral, and socioeconomic risk factors. This study used the CDC Diabetes Health Indicators dataset to evaluate diabetes risk through predictive modeling, fairness evaluation, and propensity score matching. Logistic Regression, Random Forest, and XGBoost models were trained to predict diabetes status using demographic, clinical, behavioral, and self-reported health indicators. Because diabetes-positive cases were the minority class, evaluation emphasized PR-AUC, recall, precision, false negative rate, and calibration. XGBoost achieved the strongest overall performance, including the highest PR-AUC, recall, and the lowest Brier score. Fairness analysis showed that aggregate performance masked subgroup disparities, particularly across age, income, and education groups. Propensity score matching was then used to examine the association between physical activity and diabetes after reducing observed confounding. Physical activity remained statistically associated with lower odds of diabetes, but the effect size was modest, suggesting that much of the raw association was explained by differences in BMI, health status, and socioeconomic conditions. These findings highlight the importance of integrating prediction, fairness assessment, and causal adjustment when analyzing diabetes risk in observational health data.

Keywords— Diabetes Risk Prediction; Machine Learning; Fairness Evaluation; Propensity Score Matching; Physical Activity

I. INTRODUCTION

Diabetes mellitus has become one of the most serious public health concerns in recent decades. According to the Centers for Disease Control and Prevention, more than 40 million Americans are living with diabetes, and type 2 diabetes accounts for approximately 90 to 95 percent of diagnosed cases [1]. Diabetes is strongly associated with modifiable lifestyle and clinical risk factors, including obesity, diet, physical inactivity, hypertension, and cholesterol status. Long-term complications of uncontrolled diabetes include end-stage renal disease, cardiovascular disease, neuropathy, and other microvascular and macrovascular complications, creating substantial burden for patients and healthcare systems [2].

Early identification of individuals at elevated diabetes risk is clinically important because lifestyle modification, screening, and timely intervention may delay disease progression and reduce long-term complications. Machine learning has become increasingly useful for diabetes risk prediction because it can integrate demographic, behavioral, clinical, and self-reported health indicators within a single predictive framework. Prior work using Behavioral Risk Factor Surveillance System data has shown that survey-based machine learning models can provide useful tools for initial type 2 diabetes screening and early intervention [3]. However, in imbalanced healthcare

datasets, overall accuracy or discrimination alone may be insufficient, since models can appear strong overall while still missing a substantial portion of diabetes-positive individuals.

In addition to predictive performance, fairness evaluation is an important consideration for responsible health data science. A diabetes risk model may perform well on average but still show unequal sensitivity across demographic or socioeconomic subgroups. Such disparities are clinically meaningful in screening settings because false negatives may delay identification of high-risk individuals and reduce opportunities for prevention. Recent diabetes-related machine learning studies have shown that models with reasonable aggregate performance can still exhibit subgroup bias, reinforcing the need to evaluate model performance across population groups [4].

Physical activity has also been widely studied as a protective factor against type 2 diabetes. Prior work using propensity score matching has suggested that regular physical activity is associated with improved diabetes-related outcomes after adjustment for confounding [5, 6]. However, physically active individuals may differ systematically from inactive individuals in body mass index, general health, income, education, healthcare access, and other lifestyle factors. Therefore, naive comparisons may overestimate the independent association between physical activity and diabetes. Propensity score matching provides one approach to reduce observed confounding and compare physically active and inactive individuals with more similar baseline characteristics.

This study addresses three related clinical and public

health questions. First, can commonly collected demographic, behavioral, clinical, and self-reported health indicators be used to identify adults at elevated risk of diabetes? Second, does the predictive model detect diabetes-positive individuals equally well across age, sex, income, and education subgroups? Third, after balancing observed clinical, behavioral, and socioeconomic differences through propensity score matching, is physical activity still associated with lower diabetes risk?

By addressing these questions together, this study provides a more complete view of diabetes risk in observational health data. The goal is not only to build an accurate predictive model, but also to evaluate whether model errors are distributed unevenly across subgroups and whether an observed lifestyle association remains after adjustment for confounding. This integrated framework combines predictive modeling, subgroup fairness evaluation, and causal-style analysis to support more responsible interpretation of diabetes risk prediction results.

II. MATERIALS AND METHODS

a. Data Set

This study used the CDC Diabetes Health Indicators dataset obtained from the UCI Machine Learning Repository [7]. The dataset consists of cross-sectional survey data from U.S. adults and includes demographic, behavioral, and health-related variables relevant to diabetes risk. Key variables include body mass index (BMI), blood pressure status, cholesterol status, smoking behavior, physical activity, general health, and socioeconomic indicators such as income and education.

The primary outcome variable was *Diabetes_binary*, indicating whether an individual had been diagnosed with diabetes or prediabetes. The key exposure variable for causal analysis was *PhysActivity*, a binary indicator of whether an individual reported engaging in physical activity. Most variables were represented as binary or ordinal measures, making the dataset suitable for both classification modeling and propensity score matching.

b. Study Population

The study population included individuals with complete data on the outcome, exposure, and relevant covariates. Observations with missing values in key variables were excluded to maintain data quality and analytical consistency. Individuals were categorized into two groups based on physical activity status: those who reported physical activity and those who did not.

The analysis was structured into three components: predictive modeling for diabetes risk, fairness evaluation across socioeconomic and demographic subgroups, and causal analysis using propensity score matching. The predictive modeling component trained logistic regression, random forest, and XGBoost models. The fairness analysis evaluated subgroup performance using PR-AUC, recall, precision, and false negative rate (FNR). The causal analysis used propensity score matching to reduce observed confounding and construct a matched sample of individuals with and without physical activity.

c. Predictive Modeling

Before model training, we examined the distribution of the binary outcome. The dataset was imbalanced, with approximately 86% of observations belonging to the non-diabetic class and 14% belonging to the diabetic class. Because this imbalance can bias models toward the majority class, model evaluation emphasized metrics that are more informative for minority-class detection, including PR-AUC, recall, precision, and FNR.

Prior studies have shown that survey-based health indicators can be used to build type 2 diabetes risk prediction models with machine learning methods, including logistic regression, random forest, decision trees, support vector machines, and neural networks [3]. Because diabetes-positive cases represented a minority of the dataset, PR-AUC was emphasized in addition to ROC AUC, as precision-recall curves are more informative than ROC curves for evaluating binary classifiers under class imbalance [8]. To predict diabetes risk, three supervised machine learning models were implemented: Logistic Regression, Random Forest, and XGBoost. Logistic Regression was used as a baseline model due to its interpretability and widespread application in healthcare studies, while Random Forest and XGBoost were selected to capture nonlinear relationships and interactions among variables.

The dataset was split into training and testing sets using a 70/30 ratio with stratification to preserve the class distribution. Feature scaling (standardization) was applied for Logistic Regression to ensure that variables with larger magnitudes do not dominate the model. In addition, class weights were set to “balanced” to address the class imbalance between diabetic and non-diabetic observations.

Model performance was evaluated using multiple metrics, including Accuracy, Recall, Precision, F1-score, ROC AUC, PR AUC, and Brier Score. ROC AUC was used to assess overall model discrimination, while PR AUC was emphasized due to the imbalanced nature of the dataset, as it better reflects performance on the minority class. Brier Score was used to evaluate the calibration of predicted probabilities.

For tree-based models (Random Forest and XGBoost), Threshold tuning was performed to improve recall for the diabetic class. Instead of using the default threshold of 0.5, lower thresholds were tested, and a value of 0.15 was selected based on improved recall performance. This adjustment allows the models to better identify individuals at risk of diabetes, which is particularly important in a healthcare context.

d. Fairness Evaluation

In healthcare applications, predictive models should be evaluated not only by aggregate performance but also by subgroup performance, since models may reproduce or amplify disparities across demographic and socioeconomic groups [9]. Therefore, we evaluated recall and false negative rate across sex, age, income, and education subgroups. Fairness was evaluated as a subgroup performance analysis of the final XGBoost model on the held-out test set from the same 70/30 stratified split used in modeling. We used the

same hyperparameters and the same threshold of 0.15, selected to prioritize recall in this screening-oriented task. Subgroups were defined by sex, age, income, and education, using collapsed categories to improve interpretability. Performance was assessed with PR-AUC, recall, precision and FNR, with recall and FNR treated as the main fairness metrics. We also computed subgroup gaps relative to overall test-set performance.

e. Propensity Score Matching

Propensity score matching was used to reduce observed confounding by balancing measured covariates between physically active and inactive individuals, following the framework introduced by Rosenbaum and Rubin [10]. Similar PSM approaches have been used to evaluate associations between physical activity and diabetes-related or chronic disease outcomes in observational data [5, 6]. Propensity scores were estimated using logistic regression, adjusting for demographic, clinical, behavioral, and socioeconomic covariates, including BMI, hypertension, cholesterol status, smoking, general health, age, education, and income.

We performed 1:1 nearest-neighbor matching with replacement based on estimated propensity scores. A caliper was used to avoid poor matches, and covariate balance was assessed using standardized mean differences (SMDs). SMD values below 0.1 were considered evidence of adequate balance. After matching, the association between physical activity and diabetes was estimated using logistic regression on the matched sample and reported as an odds ratio with a 95% confidence interval.

III. EXPLORATORY DATA ANALYSIS

Exploratory data analysis was used to identify the main structure of diabetes risk before modeling. The analytic dataset contained 253,680 respondents and 22 health indicator variables, with diabetes coded as a binary outcome. Diabetes-positive respondents made up 13.9% of the sample, indicating that later model evaluation should not rely only on overall accuracy.

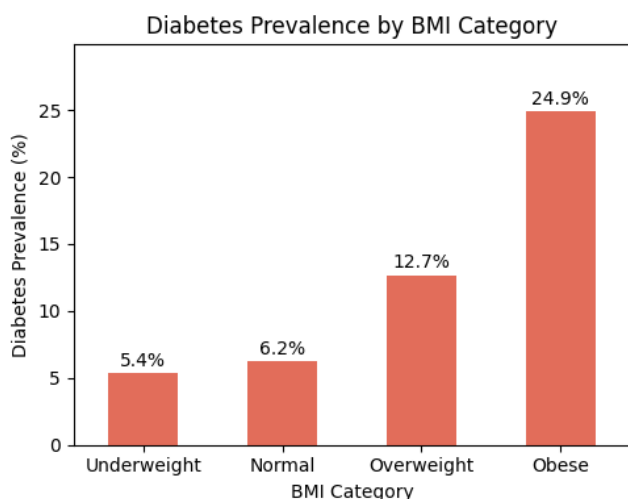


Fig. 1: BMI category prevalence

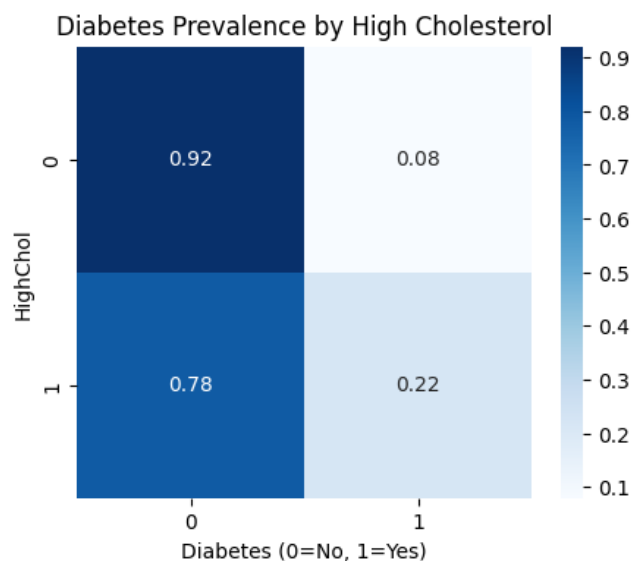


Fig. 2: High cholesterol prevalence

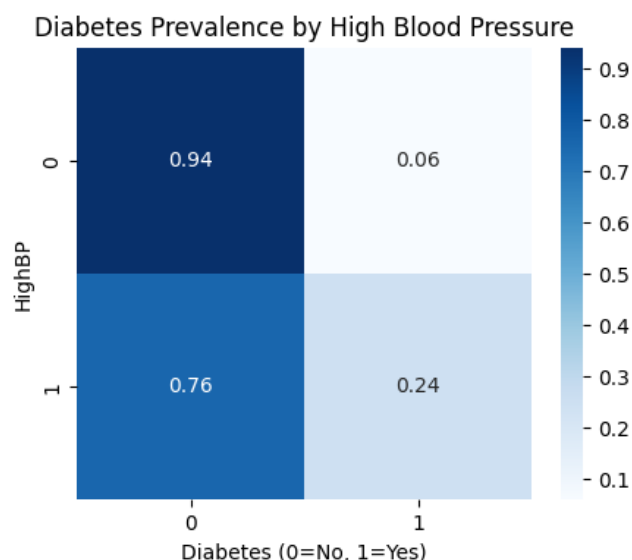


Fig. 3: High blood pressure prevalence

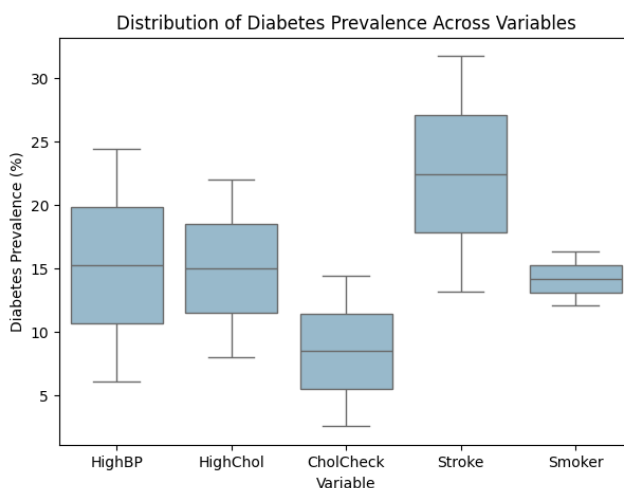


Fig. 4: Distribution across selected variables

Clinical and behavioral variables reveal consistent

patterns in diabetes risk across multiple factors. In Figure 1, diabetes prevalence increases markedly across BMI categories, rising from 5.4% in the underweight group to 24.9% among obese individuals, indicating a strong positive association between body mass and diabetes. Figures 2 and 3 further show that individuals with high blood pressure and high cholesterol have substantially higher diabetes prevalence compared to those without these conditions (24% vs. 6% for high blood pressure, and 22% vs. 8% for high cholesterol), suggesting that cardiometabolic risk factors tend to cluster together. In Figure 4, the distribution of diabetes prevalence across variables indicates that stroke, high blood pressure, and high cholesterol are associated with higher median prevalence and greater variability, while factors such as cholesterol check show relatively lower levels. These consistent gradients suggest that diabetes risk is influenced by overlapping clinical conditions, motivating the use of predictive models that can capture complex relationships and the need for further fairness analysis.

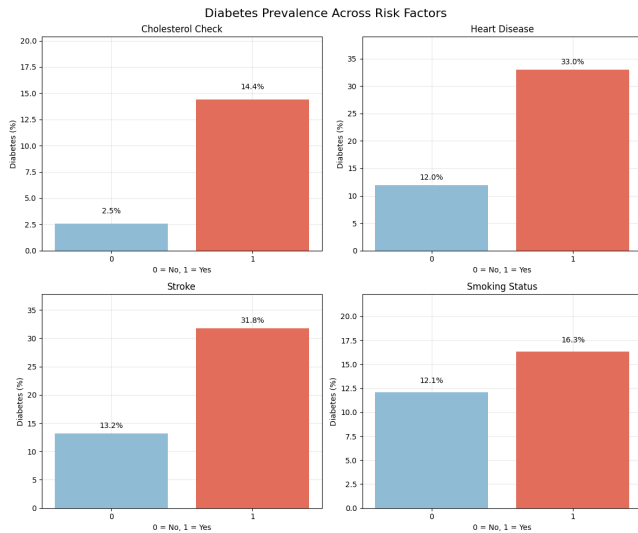


Fig. 5: Selected risk-factor prevalence panels.

This figure illustrates the relationship between four key health risk factors (Cholesterol Check, Heart Disease, Stroke, and Smoking) and diabetes prevalence. Overall, a consistent pattern is observed: individuals with these risk factors tend to have substantially higher diabetes prevalence.

For Cholesterol Check, individuals who reported having a cholesterol check show a diabetes prevalence of 14.4%, compared to only 2.5% among those who did not. This likely reflects underlying health conditions, as individuals undergoing screening may already be at higher risk.

In terms of Heart Disease, the difference is much more pronounced. Individuals with heart disease have a diabetes prevalence of 33.0%, compared to 12.0% among those without, indicating a strong comorbidity between cardiovascular disease and diabetes. A similar pattern is observed for Stroke, where individuals with a history of stroke have a prevalence of 31.8%, compared to 13.2% for those without. This further supports the close association between major cardiovascular conditions and diabetes. For smoking Status, smokers show a diabetes prevalence of 16.3%, slightly higher than 12.1% among non-smokers. While the difference is smaller compared to other factors.

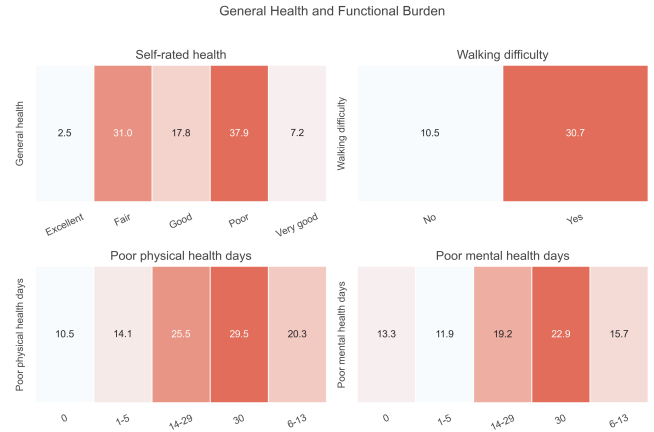


Fig. 6: Diabetes prevalence by general health and functional burden categories

General health and demographic variables added a second layer to the risk profile. Diabetes prevalence increased from 2.5% among respondents reporting excellent general health to 37.9% among those reporting poor general health. Walking difficulty and poor physical health days also showed clear burden patterns, while poor mental health days showed a weaker but nonzero signal. Diabetes prevalence varied strongly by age and socioeconomic position: prevalence increased from 1.4% among respondents aged 18–24 to 21.8% among those aged 70–74, and declined from 24.3% in the lowest income group to 8.0% in the highest income group. College graduates also had lower prevalence than lower-education groups.

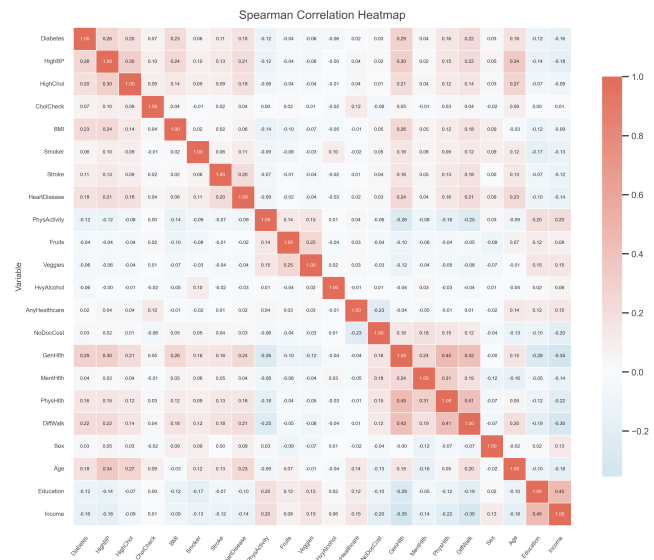


Fig. 7: Spearman correlation matrix for all variables.

The correlation heatmap showed interpretable blocks among predictors: cardiometabolic variables clustered together, general health and functional limitation formed a health-burden block, and income and education formed a socioeconomic block. This structure cautions against interpreting any single EDA association as causal. Instead, diabetes status appears linked to overlapping domains of metabolic burden, functional health, and social disadvantage.

IV. MODELING RESULTS

a. Model Selection

Three classification models were implemented to predict diabetes risk: Logistic Regression, Random Forest, and XGBoost. Logistic Regression was selected as a baseline model due to its simplicity, interpretability, and widespread use in clinical prediction tasks. Random Forest and XGBoost were chosen as ensemble learning methods capable of capturing nonlinear relationships and interactions among variables that a linear model may not detect. Together, these models provide a comparison between interpretable and more flexible predictive approaches.

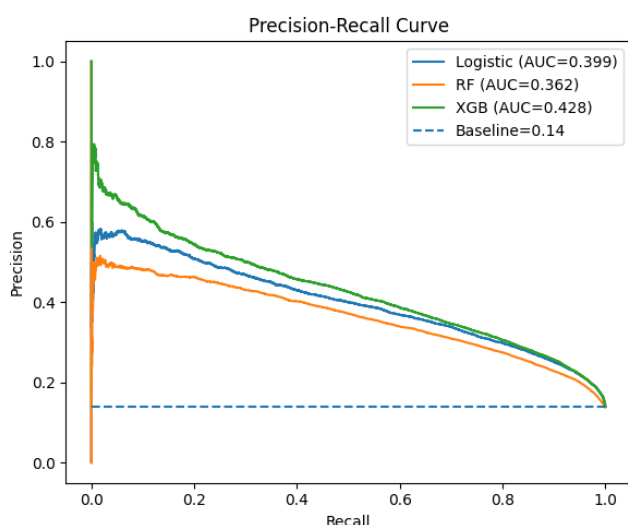


Fig. 8: Precision-recall curve for model comparison.

b. Logistic Regression

Logistic Regression was trained with balanced class weights, and the default classification threshold of 0.5 was retained. The model achieved a recall of approximately 0.76 for the diabetic class, indicating strong sensitivity in identifying positive cases without requiring additional threshold adjustment. In terms of performance, Logistic Regression achieved a ROC AUC of approximately 0.82, indicating good overall discrimination between classes. The PR AUC was 0.399, which is substantially higher than the baseline positive rate (0.14), demonstrating meaningful predictive performance under class imbalance. However, the precision remained relatively low (approximately 0.31), reflecting the trade-off between identifying more diabetic individuals and increasing false positives.

c. Random Forest

A Random Forest classifier was trained with 200 estimators and balanced class weights. When evaluated using the default classification threshold of 0.5, the model achieved a recall of only 0.15 for the diabetic class, indicating that most true positive cases were not identified. This behavior is typical in imbalanced datasets, where the default threshold favors the majority class. To improve detection of diabetic cases, threshold tuning was performed by evaluating thresholds between 0.4 and 0.1. A threshold of 0.15 was selected, as it increased recall to approximately 0.75,

representing a substantial improvement in sensitivity. The Random Forest model achieved a ROC AUC of approximately 0.79 and a PR AUC of 0.362, indicating moderate predictive performance. Despite the improvement in recall, precision remained relatively low, highlighting the trade-off inherent in imbalanced classification problems.

d. XGBoost

An XGBoost classifier was trained with 200 estimators, a maximum tree depth of 4, and a learning rate of 0.05. Similar to Random Forest, the default threshold resulted in low recall for the diabetic class. Therefore, threshold tuning was applied using the same approach, and a threshold of 0.15 was selected. With this adjustment, recall increased to approximately 0.77, slightly outperforming both Logistic Regression and Random Forest in identifying diabetic cases. XGBoost achieved the highest ROC AUC (around 0.83) and PR AUC (0.428) among all models, indicating superior performance, particularly for the minority class.

e. Model Comparison and Results

The performance of all three models is summarized in Table 1. While ROC AUC values suggest that all models perform well overall, this metric can be overly optimistic in imbalanced datasets because it evaluates performance across all thresholds and is less sensitive to minority class detection. In contrast, PR AUC provides a more informative evaluation by focusing on the model's ability to identify diabetic cases. All models achieved PR AUC values substantially higher than the baseline (0.14), indicating that they capture meaningful patterns in the data. Among them, XGBoost achieved the highest PR AUC (0.428), followed by Logistic Regression (0.399) and Random Forest (0.362).

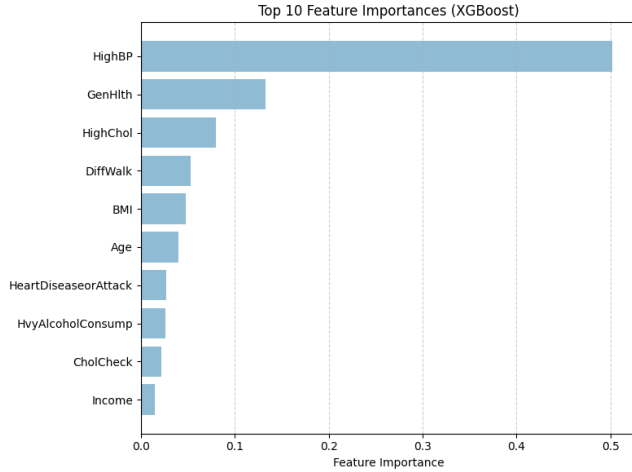
Across all models, precision remained relatively low (approximately 0.29–0.32). This is expected in imbalanced classification tasks, where improving recall typically results in more false positives. In a clinical screening context, this trade-off is acceptable, as failing to identify a high-risk individual carries greater consequences than generating additional false alarms.

In terms of probability calibration, XGBoost achieved the lowest Brier score (0.097), followed by Random Forest (0.104), while Logistic Regression had the highest Brier score (0.177). This suggests that ensemble models provide better-calibrated probability estimates.

Overall, XGBoost demonstrated the strongest performance across most evaluation metrics, including ROC AUC, PR AUC, recall, and calibration. Logistic Regression remains competitive and offers the advantage of interpretability, while Random Forest showed slightly weaker performance compared to the other models.

TABLE 1: MODEL PERFORMANCE COMPARISON ACROSS MULTIPLE EVALUATION METRICS.

Model	Accuracy	ROC AUC	PR AUC	Recall	Precision	F1-score	Brier
Logistic Regression	0.732	0.822	0.399	0.762	0.311	0.442	0.177
Random Forest	0.713	0.798	0.362	0.750	0.293	0.422	0.104
XGBoost	0.738	0.829	0.428	0.772	0.319	0.451	0.097

**Fig. 9:** Top 10 feature importance from the XGBoost model. High Blood Pressure, general health, and high cholesterol are the most influential predictors of diabetes risk.

The feature importance analysis from the XGBoost model highlights the relative contribution of each predictor to diabetes risk prediction. As shown in Figure 9, HighBP (High Blood Pressure) is by far the most influential variable, indicating that blood pressure status plays a dominant role in the model’s decision-making process. This is followed by GenHlth (self-reported general health) and HighChol (high cholesterol), both of which are key indicators of overall health condition and cardiometabolic risk. Other variables such as DiffWalk (difficulty walking), BMI, and Age also contribute meaningfully, reflecting the importance of physical health, body composition, and aging in diabetes risk. In contrast, variables such as Heart Disease or Attack, HvyAlcoholConsump, CholCheck, and Income show relatively lower importance, suggesting that while they are associated with diabetes, their predictive contribution is less significant when considered alongside other factors. Overall, the results indicate that cardiometabolic and general health-related variables dominate the predictive signal, which is consistent with the patterns observed in the exploratory data analysis. This alignment between model-driven importance and EDA findings strengthens the validity of the model and supports its interpretability.

V. FAIRNESS RESULTS

Fairness analysis evaluated subgroup performance disparities for the final XGBoost model on the held-out test set. The analysis used the same split, parameters, and decision threshold as the main modeling notebook. The threshold was 0.15, chosen to prioritize sensitivity for the diabetes-positive class. Because diabetes was the minority outcome, the main fairness metrics were PRAUC, recall, and false negative rate (FNR). Overall, the model achieved PRAUC 0.428, recall 0.773, precision 0.318, and FNR

0.227, which is more consistent with a screening model than a high-precision diagnostic model.

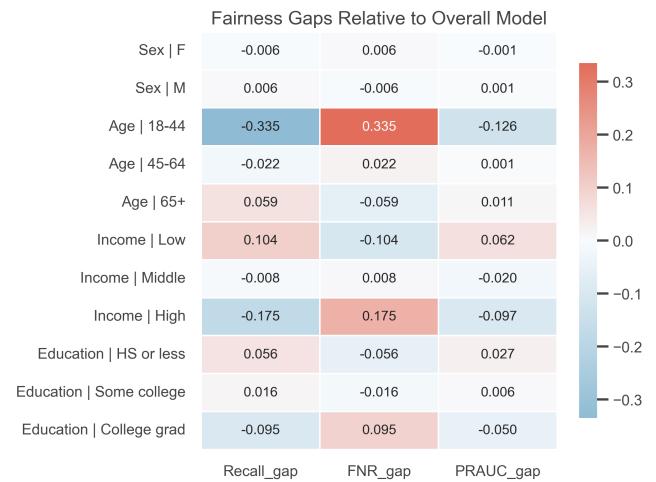
**Fig. 10:** Fairness gap heatmap comparing subgroup performance to overall model performance.

Figure 10 and Table 2 show that sex was not the main source of subgroup disparity: female and male respondents had similar recall and FNR. The clearest disparity appeared by age. The 18-44 group had recall 0.438 and FNR 0.562, compared with overall recall 0.773 and FNR 0.227, making it the subgroup most likely to have diabetes cases missed by the model.

Income and education showed a related pattern. High-income respondents had lower sensitivity than low-income respondents (recall 0.598 vs 0.877), and college graduates had lower sensitivity than respondents with high school education or less (recall 0.678 vs 0.829). A more defensible interpretation is subgroup performance disparity: the model detects diabetes more readily in groups where diabetes prevalence and risk-factor concentration are higher, while missing more cases in lower-prevalence or less typical subgroups.

TABLE 2: SUBGROUP PERFORMANCE OF THE FINAL XGBOOST MODEL.

Subgroup	PR-AUC	Recall	FNR	Key interpretation
Overall	0.428	0.773	0.227	Reference performance
Female	0.427	0.767	0.233	Similar to overall
Male	0.429	0.779	0.221	Similar to overall
Age 18–44	0.302	0.438	0.562	Largest false-negative burden
Age 65+	0.439	0.832	0.168	Higher sensitivity than overall
Low income	0.490	0.877	0.123	More readily detected by model
High income	0.331	0.598	0.402	Lower sensitivity than overall
High school or less	0.455	0.829	0.171	Higher sensitivity than overall
College graduate	0.378	0.678	0.322	Lower sensitivity than overall

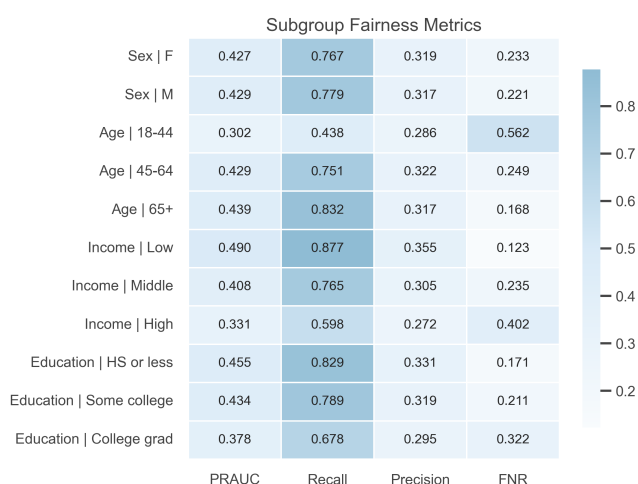


Fig. 11: Fairness heatmap for subgroup PR-AUC, recall, precision, and FNR

Figure 11 helps separate ranking behavior from threshold behavior. Some groups with weaker recall are not uniformly poor in precision, suggesting that the main issue is not simple prediction failure but sensitivity under a fixed threshold. Future work should examine threshold sensitivity, subgroup calibration, and class weighting to test whether FNR gaps can be reduced without an unacceptable increase in false positives.

VI. PROPENSITY SCORE MATCHING RESULTS

Prior to matching, individuals who engaged in physical activity differed substantially from those who did not across multiple observed characteristics, indicating selection bias and confounding. The distributions of propensity scores for physically active and inactive individuals were separated in the original sample.

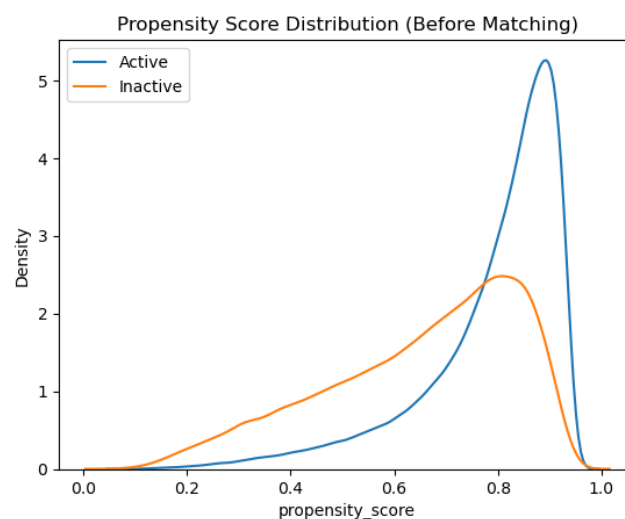


Fig. 12: Propensity score distribution before matching. Separation between active and inactive groups suggests baseline differences and potential selection bias.

After applying 1:1 nearest-neighbor propensity score matching, the distributions of propensity scores between the two groups overlapped substantially, suggesting improved comparability.

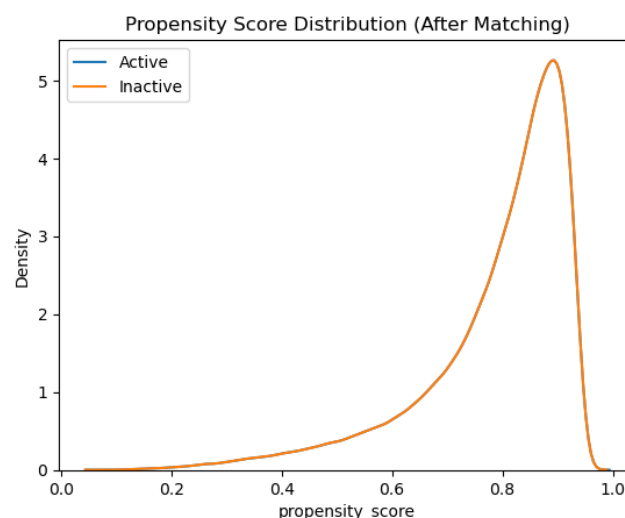


Fig. 13: Propensity score distribution after matching. Increased overlap indicates improved comparability after matching.

This improvement in balance was further confirmed by standardized mean differences. Before matching, several covariates exhibited notable imbalance with $|SMD| > 0.1$.

After matching, all covariates achieved acceptable balance with $|SMD| < 0.1$, indicating that matching effectively reduced observed differences between groups.

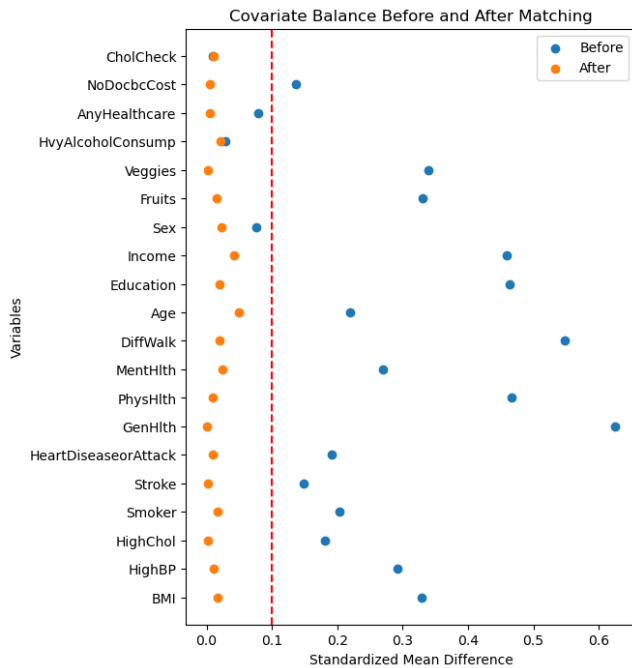


Fig. 14: Covariate balance before and after matching. Standardized mean differences were reduced after matching, with all covariates achieving acceptable balance.

In the original unmatched dataset, individuals who engaged in physical activity had a substantially lower prevalence of diabetes than those who did not. The difference in diabetes prevalence between the groups was approximately 9.5 percentage points. After matching, this difference was reduced to approximately 0.3–0.5 percentage points. This attenuation suggests that much of the raw association was driven by confounding factors, including differences in health status, lifestyle, and socioeconomic conditions.

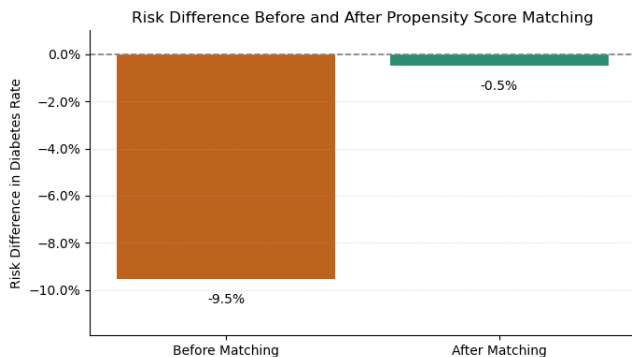


Fig. 15: Risk difference before and after propensity score matching.

Logistic regression conducted on the matched sample showed that physical activity remained significantly associated with diabetes status. Individuals who engaged in physical activity had lower odds of diabetes compared with those who did not (OR = 0.953, 95% CI: 0.935–0.972, $p < 0.001$).

TABLE 3: LOGISTIC REGRESSION RESULTS AFTER PROPENSITY SCORE MATCHING.

Variable	Odds Ratio	95% CI	p-value
PhysActivity	0.953	0.935–0.972	< 0.001

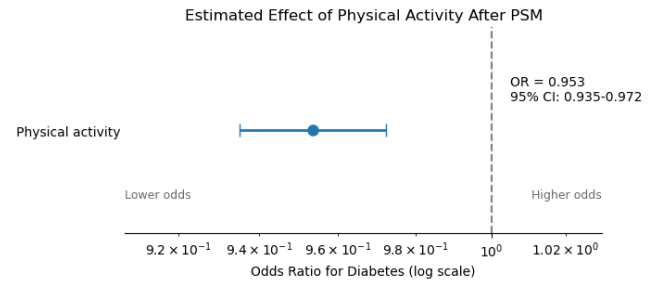


Fig. 16: Odds ratio for the association between physical activity and diabetes after propensity score matching.

Although the association was statistically significant, the magnitude of the effect was modest. After taking into account the observed confounders, the independent contribution of physical activity to the risk of diabetes appeared relatively small. The comparison between unmatched and matched results highlights the importance of adjusting for confounding in observational data.

VII. CONCLUSIONS

This study mainly integrated three analytical component - predictive modeling, fairness evaluation, and causal inference, to provide a more complete understanding of diabetes risk using the CDC Diabetes Health Indicators dataset.

For predictive modeling, all three models achieved reasonable discrimination, with ROC-AUC values ranging from 0.798 to 0.829. However, PR-AUC was a more informative metric given the class imbalance, with XGBoost performing best at 0.428. Threshold tuning was necessary to achieve clinically acceptable recall for the diabetic minority class, highlighting that default model settings are insufficient in imbalanced healthcare data.

Fairness analysis revealed that overall model performance varied largely between all subgroups. While sex differences were small, younger adults aged 18 to 44 showed the largest false negative burden and high-income respondents and college graduates also had lower recall, suggesting the model tends to miss diabetes cases in subgroups where the disease is less common.

The propensity score matching analysis showed that physical activity was statistically significant associated with reduced diabetes risk after controlling for observed confounders (OR = 0.953, 95% CI: 0.935–0.972, $p < 0.001$). However, the effect size was modest, and the raw risk difference 9.5 percentage points was reduced to approximately 0.3–0.5 percentage points after matching, showing that much of the original association was driven by confounding factors including BMI, general health status, and socioeconomic conditions.

Together, these findings suggest that while machine learning models can identify diabetes risk with reasonable

accuracy, their clinical application requires careful attention to subgroup fairness and metric selection. Physical activity was still significantly associated with lower diabetes risk, but the effect was small after controlling for confounding factors.

VIII. DISCUSSION

The predictive modeling results highlight several important considerations for applying machine learning to imbalanced clinical data. While all three models achieved reasonable discrimination, the gap between ROC-AUC and PR-AUC values reveals that standard metrics can be misleading when the positive class is rare [8]. A model that appears strong based on ROC-AUC alone may still miss a substantial proportion of diabetic individuals, which is precisely the outcome a screening tool should avoid [3]. The decision to lower the classification threshold to 0.15 for tree-based models was therefore a clinically motivated choice to prioritize sensitivity over precision in a population screening context.

The feature importance analysis from XGBoost further showed that high blood pressure, general health status, and high cholesterol were the dominant predictors, consistent with established clinical knowledge about cardio metabolic risk [2] and aligned with EDA findings. Notably, income ranked low in feature importance despite its strong association with diabetes prevalence in the EDA, likely because its effect is absorbed by correlated variables such as BMI and general health once included together [4]. This is a reminder that feature importance should not be interpreted as a measure of causal effect.

The fairness analysis showed that the final XGBoost model did not perform uniformly across subgroups, even though its overall screening performance was acceptable. Sex differences were small, but younger adults, high-income respondents, and college graduates had lower recall and higher false negative rates than the overall population. A plausible explanation is that diabetes cases in these lower-prevalence groups may be less typical of the risk patterns learned by the model, so a single global threshold of 0.15 does not provide equal sensitivity across subgroups. These findings suggest that aggregate metrics alone can mask clinically important subgroup disparities, and that recall and false negative rate should be reported alongside overall PR AUC.

The causal analysis showed that physical activity was associated with lower odds of diabetes after propensity score matching, but the effect size was small. In the matched sample, diabetes prevalence was 12.11% among inactive individuals and 11.61% among active individuals. This suggests that much of the raw association between physical activity and diabetes was explained by confounding.

Several limitations should be acknowledged. First, the dataset is cross-sectional, which prevents any inference about temporal ordering or disease progression. Second, PSM can only control for measured confounders, and residual confounding from unmeasured variables such as diet quality or genetic factors cannot be ruled out. Third, the fairness analysis relied on a fixed classification threshold of 0.15 applied uniformly across subgroups. Future work

should explore subgroup-specific threshold calibration to determine whether false negative rate gaps can be reduced without an unacceptable increase in false positives.

Future analyses could also consider longitudinal data to better capture the temporal relationship between physical activity and diabetes onset. Incorporating additional socioeconomic or behavioral variables, such as dietary patterns or healthcare access, could further improve both predictive accuracy and fairness across subgroups. Finally, external validation on independent datasets would be necessary before deploying any of these models in a real clinical screening context.

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